BIG DATA TECHNOLOGIES

INFRASTRUCTURE HEALTH ANALYSIS FOR DIGITAL HOSPITALS USING THE HADOOP ECOSYSTEM

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TABLE OF CONTENTS

1. ABSTRACT
2. INTRODUCTION
3. EXISTING SYSTEM
4. PROPOSED SYSTEM
5. OBJECTIVES
6. LITERATURE SURVEY
7. SOFTWARE REQUIREMENTS
8. HARDWARE REQUIREMENTS
9. FUNCTIONAL REQUIREMENTS
10. NON – FUNCTIONAL REQUIREMENTS
11. SYSTEM ARCHITECTURE
12. UML DIAGRAM
13. ACTIVITY DIAGRAM
14. MODULE DESIGN

REVIEW – 2 COMMENTS ADDRESSED

•Clarification: MySQL is not our Big Data storage; it simulates the hospital's existing legacy system.
•The Pipeline Goal: The project demonstrates extracting data *from* limited legacy systems (MySQL) *into* our actual Big Data environment (Hadoop).
•Why Hadoop over MongoDB: MongoDB is an operational NoSQL database (good for live apps). We need a Data Warehouse (Hive on HDFS), which is specifically optimized for massive, historical, batch-analytical queries.

ABSTRACT

Modern hospitals generate millions of machine-generated logs every hour from Wi-Fi routers, servers, and digital medical devices such as MRI and X-ray systems. Due to their volume and velocity, these logs cannot be efficiently analyzed using traditional database systems. This project implements a Hadoop-based Big Data framework to store and analyze these digital footprints. Apache HDFS is used for scalable and fault-tolerant storage, while Apache Hive enables large-scale analytical querying. The system identifies security threats such as brute-force attacks and detects abnormal machine error patterns, helping improve both network security and operational reliability in hospital environments.

MODULE-WISE IMPLEMENTATION

- Data Ingestion Module (Apache Sqoop):

1. Executed sqoop import CLI commands to connect to the MySQL metadata.
2. Automatically mapped relational tables into HDFS directories.

- Data Processing Module (MapReduce):

1. Mapper Logic: Reads raw text line-by-line, splitting strings to extract IP addresses, timestamps, and error codes.
2. Reducer Logic: Groups the extracted keys together and counts the occurrences (e.g., grouping all "Error 505" flags).

- Analytics Module (Apache Hive):

1. Created internal tables using CREATE TABLE with defined schemas.
2. Executed SELECT ... GROUP BY HiveQL queries to detect anomalies.

RESULTS

The system successfully completed the Big Data analytical pipeline:

- **Data Migration:** Successfully ingested raw log datasets into HDFS without data loss.
- **Data Cleansing:** MapReduce effectively parsed unstructured text into clean, structured fields.
- **Security Threats Identified:** Hive queries successfully flagged repeated failed authentication attempts (Brute Force).
- **Operational Intelligence:** Error code aggregation revealed recurring hardware faults for predictive maintenance.
- **Audit Trail:** Established a highly available, tamper-proof historical log repository.

```
Time taken: 0.452 seconds, Fetched: 10 row(s)
hive> set hive.cli.print.header=true;
hive> SELECT timestamp, src_ip, dest_ip, attack_label
    > FROM hospital_network_logs
    > WHERE attack_label = 'Reconnaissance'
    > LIMIT 10;
OK
timestamp       src_ip  dest_ip attack_label
15/04/2025 02:57:22 PM  192.168.1.108   192.168.1.1     Reconnaissance
15/04/2025 03:11:29 PM  192.168.1.108   192.168.1.1     Reconnaissance
15/04/2025 03:10:19 PM  192.168.1.108   192.168.1.103   Reconnaissance
15/04/2025 03:10:59 PM  192.168.1.108   192.168.1.102   Reconnaissance
15/04/2025 02:57:39 PM  192.168.1.108   192.168.1.1     Reconnaissance
15/04/2025 02:58:45 PM  192.168.1.107   192.168.1.1     Reconnaissance
15/04/2025 03:09:47 PM  192.168.1.108   192.168.1.101   Reconnaissance
15/04/2025 03:11:52 PM  192.168.1.108   192.168.1.101   Reconnaissance
15/04/2025 03:09:19 PM  192.168.1.108   192.168.1.101   Reconnaissance
15/04/2025 03:09:10 PM  192.168.1.108   192.168.1.103   Reconnaissance
Time taken: 0.077 seconds, Fetched: 10 row(s)
hive>
```

```
hive> CREATE TABLE hospital_network_logs (
    >     flow_duration STRING,
    >     dest_ip STRING,
    >     src_ip STRING,
    >     attack_label STRING
    > )
    > ROW FORMAT DELIMITED
    > FIELDS TERMINATED BY '\t'
    > STORED AS TEXTFILE;
OK
Time taken: 2.686 seconds
hive> LOAD DATA INPATH '/hospital_data/clean_output/part-00000' INTO TABLE hospital_network_logs;
Loading data to table default.hospital_network_logs
Table default.hospital_network_logs stats: [numFiles=1, totalSize=19174223]
OK
Time taken: 0.795 seconds
hive>
```


```
hive> CREATE TABLE hospital_network_logs (
    >     flow_duration STRING,
    >     dest_ip STRING,
    >     src_ip STRING,
    >     attack_label STRING
    > )
    > ROW FORMAT DELIMITED
    > FIELDS TERMINATED BY '\t'
    > STORED AS TEXTFILE;
OK
Time taken: 2.686 seconds
hive> LOAD DATA INPATH '/hospital_data/clean_output/part-00000' INTO TABLE hospital_network_logs;
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Table default.hospital_network_logs stats: [numFiles=1, totalSize=19174223]
OK
Time taken: 0.795 seconds
hive>
```
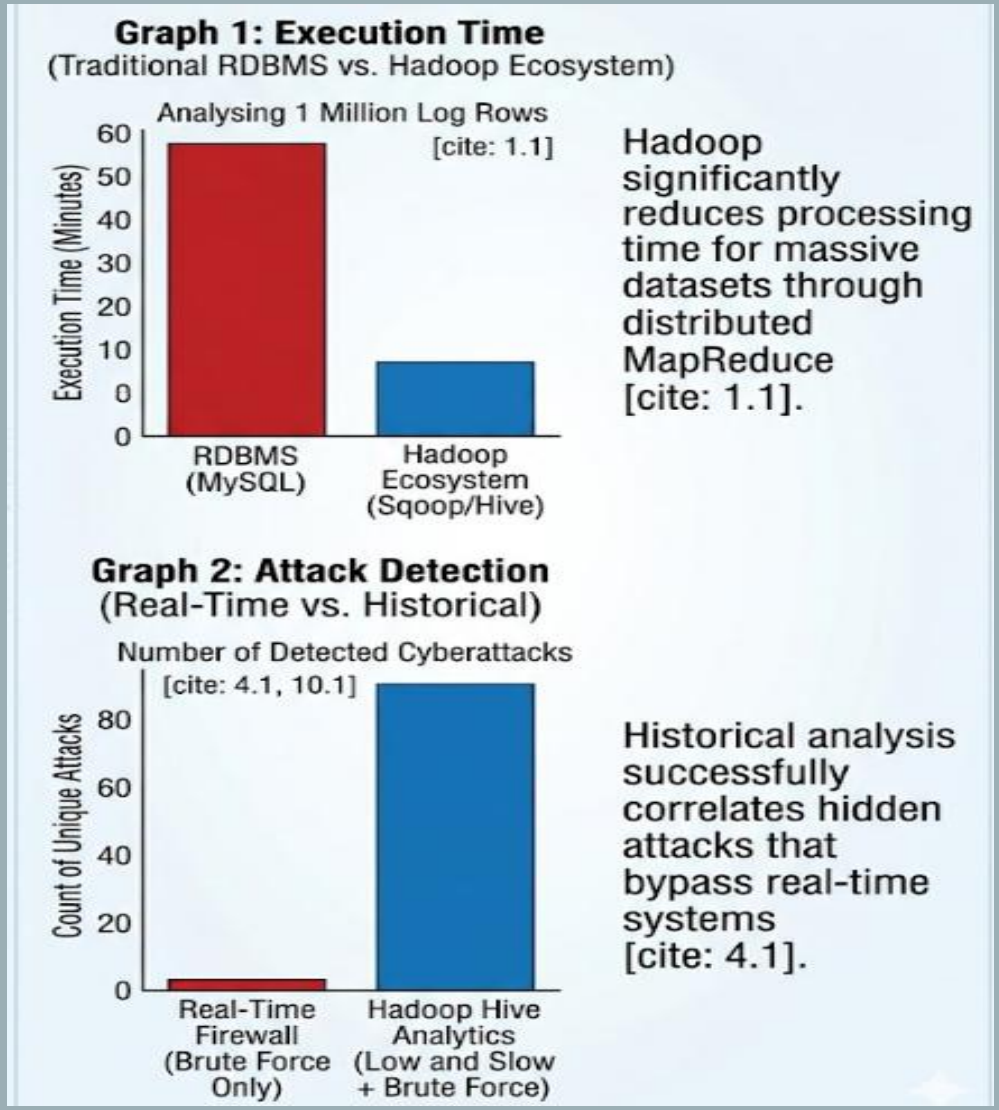

Comparison and discussion(Graphical representation)



CONCLUSION

- Traditional RDBMS and real-time firewalls are insufficient for the scale of modern hospital data.
- Implementing a Hadoop ecosystem (HDFS, MapReduce, Hive, Sqoop) solves the storage bottleneck.
- Batch processing historical logs allows for the detection of Advanced Persistent Threats (APTs) and "low and slow" attacks.
- The project provides a scalable, fault-tolerant framework to ensure both digital security and medical equipment reliability.

Future Direction

- **Real-Time Streaming:** Integrate **Apache Kafka** and **Spark Streaming** to analyze logs in real-time alongside batch processing.
- **Machine Learning (MLOps):** Apply ML clustering algorithms (like K-Means) to automatically categorize new, unknown cyberattack patterns without manual queries.
- **Cloud Migration:** Transition the local VirtualBox Cloudera cluster to a managed cloud service like AWS EMR or Google Cloud Dataproc for infinite scalability.

INTRODUCTION

The Problem:
Hospitals are high-value targets for cyberattacks such as ransomware. The massive volume of system-generated logs makes manual monitoring and analysis impractical.

The Data:
The project focuses on analyzing system logs, which are automatically generated text files produced by servers, network devices, and medical equipment.

The Solution:
A distributed log analysis framework using Hadoop. Instead of processing logs on a single machine, the system uses a cluster-based approach to detect anomalous patterns across large datasets.

Objective Overview:
To build a scalable pipeline that ingests, cleans, stores, and queries system logs to identify security threats and operational issues.

EXISTING SYSTEM

Current Method:

- Hospitals use traditional **RDBMS** (MySQL/Oracle) for log storage.
- Security relies on **Firewalls** and real-time **IDS**.
- Data retention is short (24–48 hours) due to storage limits.

Major Drawbacks:

- **Storage Limit:** Traditional databases crash or slow down under terabytes of log data.
- **Security Gap:** Firewalls miss "**Low and Slow**" attacks spread over weeks.
- **No Correlation:** Cannot link logs from Wi-Fi, Servers, and MRI machines into one timeline.
- **Reactive Maintenance:** Equipment is fixed only **after failure** because error patterns aren't analysed.

PROPOSED SYSTEM

- Our goal is to build a centralized **Data Warehouse** using Hadoop to store and analyze all hospital machine logs.
- Main purpose is to move from reactive security to **Proactive Forensic Analysis**.
- We are using **Apache Sqoop** to automate the ingestion of data from diverse sources like VPNs, Routers, and MRI machines.
- We deploy the data into **HDFS** to provide a scalable, fault-tolerant storage environment for terabytes of data.
- **Apache Hive** allows us to query 30 days of history at once to detect "low and slow" hacking patterns.
- This solution helps the hospital identify **Predictive Maintenance** needs by analyzing error trends in medical equipment.
- Utilizing this architecture will allow the hospital to maintain a **tamper-proof audit trail** for legal and regulatory compliance

OBJECTIVES

- To store and process large hospital log datasets reliably using HDFS.
- To automate data cleaning and preprocessing using MapReduce.
- To identify security threats and abnormal access patterns via Hive.
- To establish a reliable, end-to-end data pipeline using Sqoop.
- To improve operational reliability by analysing medical equipment error logs.

SOFTWARE REQUIREMENTS

Below are the software requirements for proposed system

- **Virtualization Tool:** Oracle VM VirtualBox.
- **Hadoop Environment:** Cloudera QuickStart VM (CDH - Cloudera Distribution including Hadoop).
- **Guest Operating System:** CentOS (pre-configured inside the Cloudera VM).
- **Core Framework:** Apache Hadoop (HDFS and MapReduce) for storage and processing.
- **Data Ingestion:** Apache Sqoop for importing raw datasets.
- **Data Warehouse:** Apache Hive for writing analytical queries.
- **Relational Database:** MySQL (built into Cloudera) to simulate the hospital's local database.

HARDWARE REQUIREMENTS

The hardware requirements for the proposed system include the following:

- **Processor:** Intel Core i5 / AMD Ryzen 5 (or higher).
- **RAM (Memory):** Minimum 8 GB RAM.
- **Storage Space:** Minimum 50 GB of free disk space.
- **System Architecture:** 64-bit Host Operating System (Windows / Mac / Linux).
- **Network:** Standard Ethernet or Wi-Fi adapter for internal VM networking.

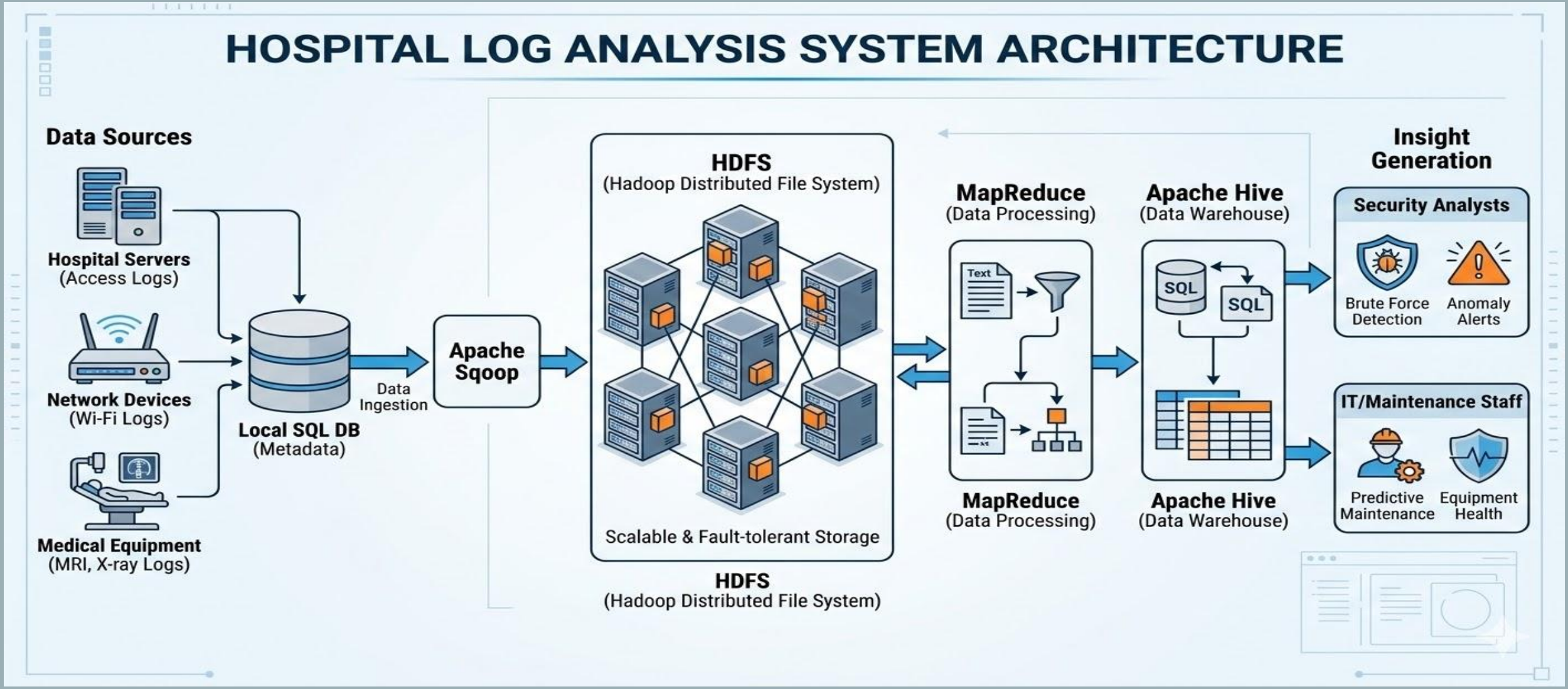
LITERATURE SURVEY

S.No	Research Paper (Title & Link)	Author(s)	Drawbacks / Gaps	Future Directions
1	**Big Data for Secure Healthcare System: A Conceptual Design** https://link.springer.com/article/10.1007/s40747-017-0040-1	Research Team (Secure Healthcare Study)	Conceptual model only; no Hadoop ecosystem implementation.	Implement full Hadoop pipeline (Sqoop → HDFS → MapReduce → Hive).
2	**Log Analysis with Hadoop MapReduce** https://dergipark.org.tr/en/pub/ject/article/980060	Gligor Risteski et al.	Uses traditional MapReduce; high latency for alert generation.	Replace batch processing with Hive/Spark SQL for faster querying.
3	**Enhanced Secured MapReduce Layer for Big Data Privacy and Security** https://doi.org/10.1186/s40537-019-0193-4	P. Jain, M. Gyanchandani, N. Khare	Focused on framework-level security; lacks practical log-analysis case study.	Integrate security layer with healthcare log analytics pipeline.
4	**Digital Forensics Readiness in Big Data Networks: A Framework for Linux–Hadoop Environments** https://doi.org/10.3390/asi7050090	Cephas Mpungu et al.	Emphasizes forensic response after attack, not proactive threat detection.	Develop proactive anomaly detection queries using Hive.
5	**A Novel Hadoop Security Model for Addressing Malicious Collusive Workers** https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8285196/	(PMC Research Group Authors)	Focused only on cluster-node security, not log-based intrusion detection.	Combine node-level protection with log anomaly detection.

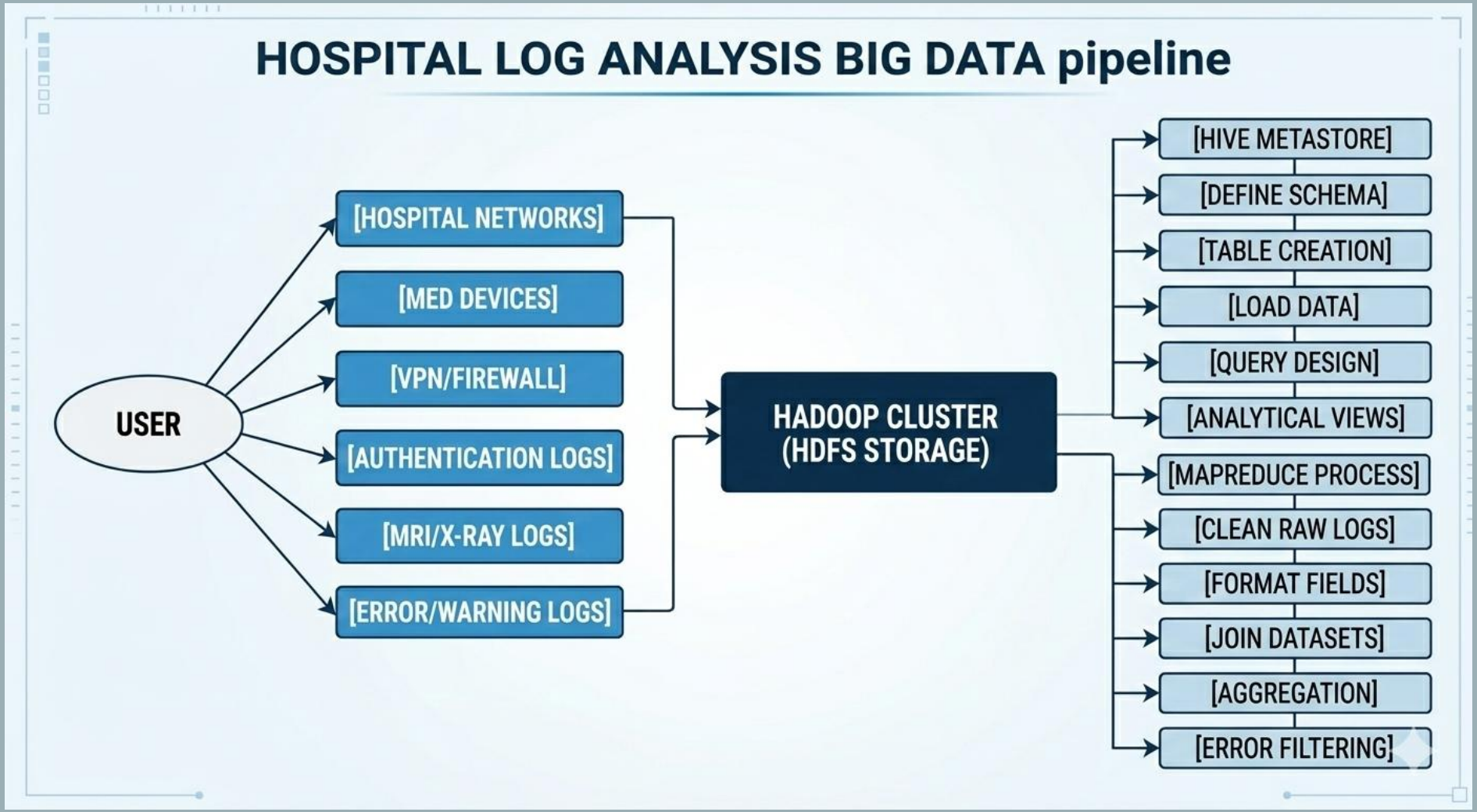
LITERATURE SURVEY

6	**Evading Cyber-Attacks on Hadoop Ecosystem: ML-Based Security-Centric Approach** https://doi.org/10.3390/info15090558	Neeraj A. Sharma et al.	Heavy ML dependency; requires high computational resources.	Implement lightweight rule-based Hive queries for smaller hospitals.
7	**Aadhaar Data Analysis Comparison in MapReduce, Hive and Spark** https://doi.org/10.2991/ahis.k.210913.036	R. Roopa et al.	Not security-focused; dataset unrelated to cybersecurity logs.	Apply comparative analysis specifically to hospital system logs.
8	**Big Data Analytics in Healthcare: Data Sources, Tools, Challenges and Opportunities** https://doi.org/10.1186/s43067-024-00190-w	Multiple Authors (Springer Open Survey)	Broad survey; lacks technical implementation details.	Develop practical Hadoop-based implementation for hospital infrastructure.
9	**Analyzing Web Access Logs using Spark with Hadoop** https://ijcaonline.org/archives/volume180/number1/28768-2017915904/	Vandita Jain et al.	Focused on web logs; not healthcare or ransomware detection.	Extend approach to medical device and network router logs.
`	**Big Data for Secure Healthcare System: A Conceptual Design** https://link.springer.com/article/10.1007/s40747-017-0040-1	Research Team (Secure Healthcare Study)	Conceptual model only; no Hadoop ecosystem implementation.	Implement full Hadoop pipeline (Sqoop → HDFS → MapReduce → Hive).

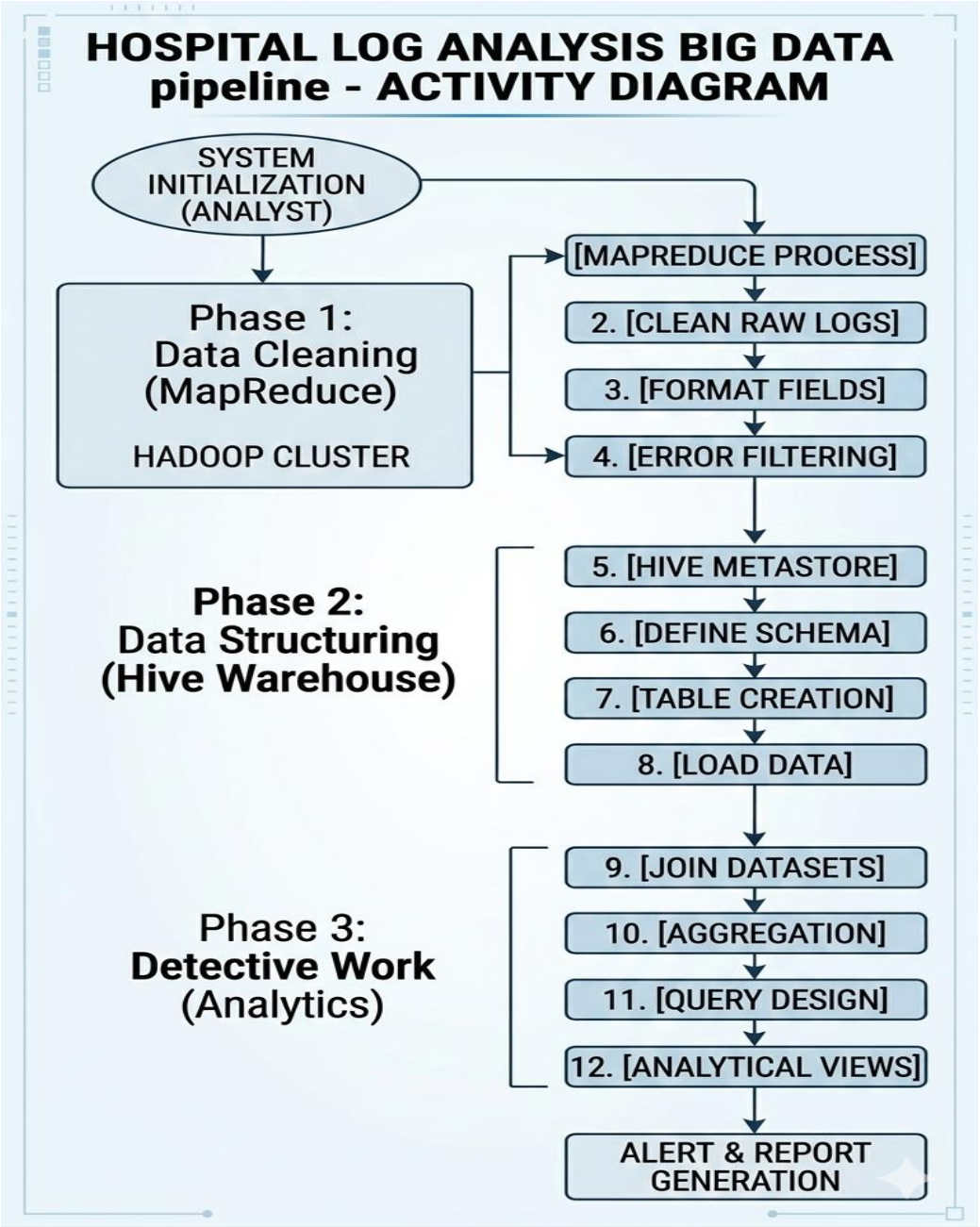
SYSTEM ARCHITECTURE



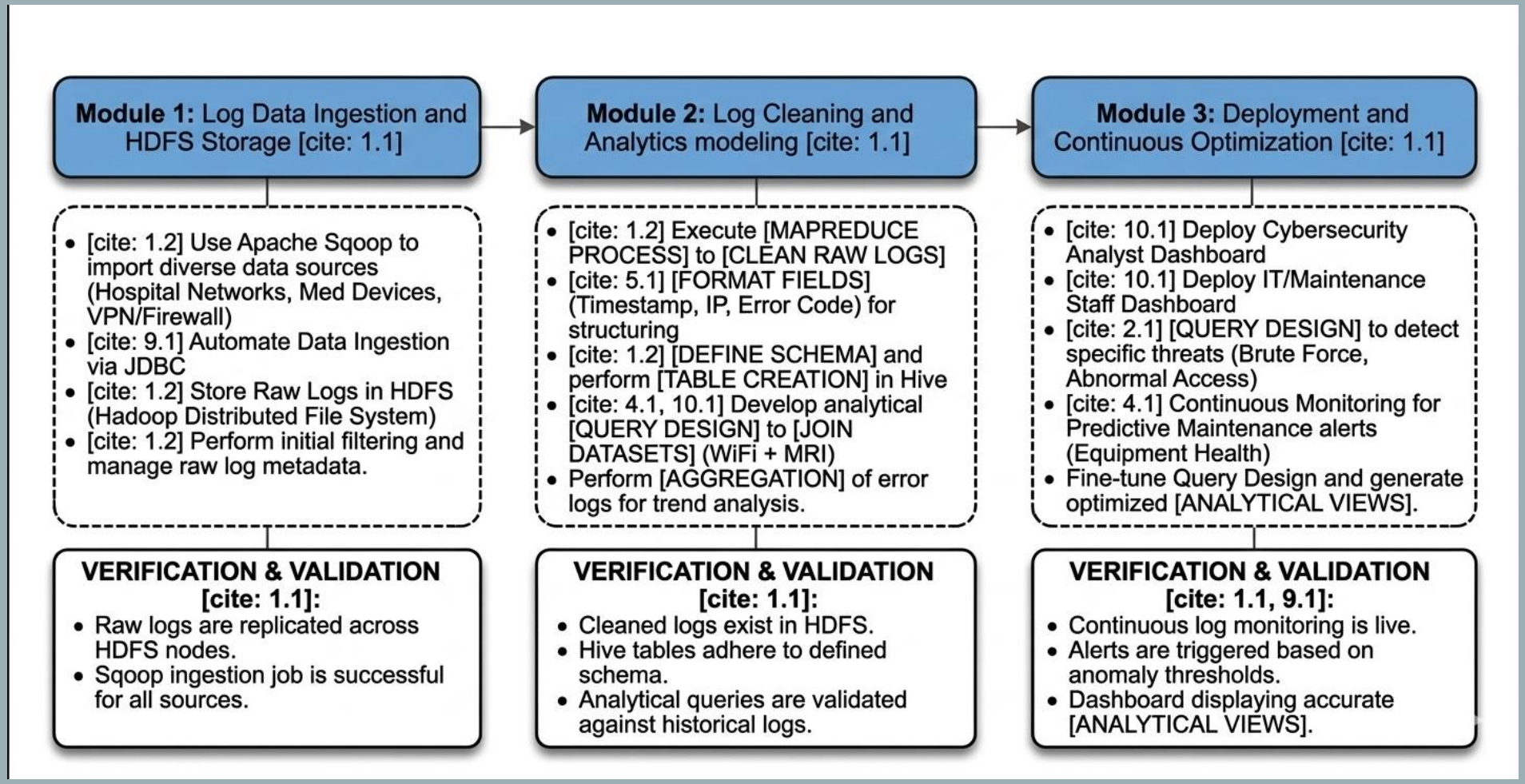
USE CASE DIAGRAM



ACTIVITY DIAGRAM



MODULE DESIGN



THANK YOU